

Machine learning researcher and engineer with deep expertise in **explainable AI**, **computer vision**, and **generative AI**. Passionate about developing **interpretable machine learning systems** that combine theoretical rigor with practical scalability. Experienced in designing multimodal AI applications across vision, language, and video domains. Interested in advancing **AI for scientific discovery** and building production-grade ML systems that are both powerful and transparent.

EDUCATION

MSc in Advanced Computing (Computer Science), *King's College London*
London, United Kingdom

Sep 2023 — Jan 2025

- Graduated with **Distinction** (among the top students in the cohort).
- **Master's Thesis**: "Real-Time Incremental Explanations and Saliency Maps for Object Detection in Videos" (92/100), supervised by *Prof. Hana Chockler*.
- **Scholarship**: Colfuturo - Foundation for the Future of Colombia (ranked 4th nationwide, top 0.5% of 779 shortlisted candidates from Colombia).
- **Relevant Coursework**: Neural Networks and Deep Learning (including **generative models such as VAEs and GANs**), Computer Vision, Data Mining, Multi-Agent Systems.
- **Extracurriculars**: Research Assistant in Explainable AI. Student Lead, Google Developer Student Club (GDSC).

BSc in Mechatronics Engineering, *Universidad Militar Nueva Granada*
Bogotá, Colombia

Jan 2017 — Apr 2022

- Graduated **Cum Laude** with the **highest GPA in cohort (4.54/5.0)**.
- **Bachelor's Thesis**: "Design and Implementation of Nonlinear Controllers for 3D Printing Systems" (5.0/5.0), awarded *Tesis Laureada* (highest honors).
- **Honors & Awards**: University Medal for Academic Merit, 5× *Matrícula de Honor* (highest semester GPA).
- **Scholarship**: Full scholarship awarded by the Colombian Ministry of Information Technologies and Communications.
- **Extracurriculars**: Contributed to the DaVinci research group on nonlinear control systems. Competitor on the university karate team (multiple tournament victories).

RESEARCH EXPERIENCE

Research Assistant - Volunteer, *King's College London*
London, United Kingdom

Sep 2023 — Present

- Developed the **IncX algorithm** for real-time explanations of object detectors, achieving an **order-of-magnitude reduction in latency** compared to benchmarks.
- Conducting research in Explainable AI with a focus on efficient and scalable **interpretation of deep neural networks**, with submissions to top-tier AI venues.
- Investigated efficient approaches to interpreting complex model behavior, with interest in extending these ideas to generative models.
- Contributed to XAI methods for computer vision (**IncX**, **ReX**), leveraging PyTorch and ONNX within reproducible ML workflows.
- **Research Areas**: Explainable AI, Computer Vision

Teaching Assistant, *Universidad Militar Nueva Granada*
Bogotá, Colombia

Aug 2020 — Jun 2021

- Delivered tutorials for the Mechatronic Systems Models module.
- Contributed to the DaVinci research group, focusing on nonlinear control systems and applied mechatronics.

PUBLICATIONS

1. **Explainable AI for the Classification of Brain MRIs**
European Conference on Artificial Intelligence (ECAI) 2025 Workshop on EXPLIMED
Blake N., Kelly D. A., Chockler H., **Calderón-Peña S.**, Chanchal A.
[ResearchSquare Preprint](#)
2. **Real-Time Incremental Explanations for Object Detectors** *[Under Review]*
Submitted
Calderón-Peña S., Chockler H., Kelly D. A.
[arXiv Preprint](#)

INDUSTRY EXPERIENCE

Software Engineer – AI

Jan 2025 — Present

Linqia (Remote, Core Engineering team based in San Francisco, USA)

- Built scalable **multimodal AI pipelines** integrating vision and language models (Vertex AI, Gemini) to process large volumes of video data for structured information extraction.
- Designed **LLM-powered AI agents** using the **ReAct (Reason + Act)** framework with **Model Context Protocol (MCP)** tools for real-time retrieval, metrics computation, and system interaction.
- Applied LLMs for **clustering** and **topic modeling** on high-volume text data, enabling automated insight generation at scale.
- Implemented **semantic search** using SentenceBERT embeddings and Pinecone vector databases; optimized production ML pipelines with Docker and Kubernetes.

Software Engineer

Dec 2022 — Dec 2024

Quorum Software, Bogotá, Colombia / London, United Kingdom

- Designed and deployed a **document classification** pipeline in PyTorch with **SHAP-based explainability**; benchmarked against fine-tuned SentenceBERT baselines on Azure ML.
- Architected a **Generative AI** application using **RAG** (Azure AI Search + GPT), serving distributed teams across three countries.

Junior Software Developer

Mar 2022 — Oct 2022

Kin + Carta, Remote (US, Argentina, Colombia)

- Built and deployed Java RESTful APIs on GCP for scalable automation pipelines.

TECHNICAL SKILLS

- **Machine Learning & Data Science:** PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, ONNX, SHAP
- **Reproducibility & MLOps:** Docker, Kubernetes, GitHub Actions, CI/CD, Azure Machine Learning
- **Cloud Platforms:** Azure, Google Cloud Platform, AWS
- **Research & Documentation:** $\LaTeX$, Mendeley
- **Version Control:** Git, GitHub

CERTIFICATIONS

Microsoft Certified DP-100: Azure Data Scientist Associate, Microsoft

Credential id: [E0BEA56FD07E8AB7](#)

Microsoft Certified AI-102: Azure AI Engineer Associate, Microsoft

Credential id: [E5960BE874975467](#)

Google Cloud Certified - Associate Cloud Engineer, Google Cloud

Credential id: [63393547](#)

LANGUAGES

- **English:** TOEFL Score: 111/120 (C2)
- **French:** DELF B2 Certified (B2)
- **Spanish:** Native Speaker
- **German:** Proficiency Level (B1)
- **Italian:** Proficiency Level (B1)

VOLUNTEERING

Student Lead

Sep 2023 — Mar 2024

Google Developer Students Club - King's College London

- Led a workshop on MLOps in Google Cloud, demonstrating how to use Vertex AI to automatically train and deploy a neural network for digit recognition using the MNIST dataset.
- Conducted a session on applying Vertex AI APIs for industry applications, including text-to-speech, speech-to-text, and face recognition.
- Presented on developing APIs with Flask for natural language processing (NLP) tasks, such as lemmatization and named entity recognition using spaCy.

Computer Skills Education Volunteer

Jul 2022 — Jun 2023

Fundación Laudes Infantis

- Taught basic robotics to children in low-income areas of Bogotá, empowering dozens of individuals with hands-on STEM skills.
- Provided foundational education in JavaScript and HTML to adolescents and adults, helping them build websites to support their entrepreneurial ventures.